

Solução do Problema grow7 - Solver HiGHS

Análise Computacional

6 de outubro de 2025

1 Informações do Problema

Informações do Problema:

- Nome: grow7
- Número de Variáveis: 301
- Número de Restrições: 140
- Inviabilidade Primal: 0.000e+00
- Inviabilidade Dual: 3.999e-13
- Valor Primal: -4.779e+07
- Valor Dual: 0.000e+00
- Gap: -1.631e-24
- Número de Iterações: 19

2 Variáveis Primais e Custos Reduzidos

Tabela 1: Variáveis primais (x) e custos reduzidos (z) do problema grow7

Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
1	1.856e+05	1	0.000e+00
2	1.185e+04	2	0.000e+00
3	6.040e+03	3	0.000e+00
4	9.092e+03	4	-1.643e-13
5	4.105e+03	5	0.000e+00
6	1.745e+04	6	0.000e+00
7	4.723e+04	7	0.000e+00
8	5.392e+03	8	0.000e+00
9	6.291e+04	9	-6.153e+01
10	7.115e+03	10	0.000e+00
11	3.213e+04	11	0.000e+00
12	5.935e+03	12	0.000e+00

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
13	3.658e+04	13	0.000e+00
14	2.953e+04	14	0.000e+00
15	6.745e+01	15	0.000e+00
16	6.351e+02	16	0.000e+00
17	9.894e+03	17	0.000e+00
18	4.609e+03	18	0.000e+00
19	1.367e+04	19	0.000e+00
20	6.730e+04	20	0.000e+00
21	0.000e+00	21	4.230e+00
22	0.000e+00	22	3.000e+00
23	0.000e+00	23	6.906e-01
24	1.191e+05	24	0.000e+00
25	8.561e+03	25	0.000e+00
26	0.000e+00	26	0.000e+00
27	4.546e+03	27	1.892e-13
28	0.000e+00	28	0.000e+00
29	0.000e+00	29	-1.155e-14
30	2.386e+04	30	0.000e+00
31	0.000e+00	31	0.000e+00
32	3.138e+04	32	0.000e+00
33	0.000e+00	33	0.000e+00
34	2.059e+04	34	1.021e-14
35	0.000e+00	35	0.000e+00
36	2.511e+04	36	0.000e+00
37	0.000e+00	37	0.000e+00
38	0.000e+00	38	0.000e+00
39	0.000e+00	39	0.000e+00
40	0.000e+00	40	4.374e-14
41	0.000e+00	41	-4.219e-15
42	7.108e+03	42	0.000e+00
43	0.000e+00	43	-8.882e-16
44	1.017e+06	44	0.000e+00
45	0.000e+00	45	-1.776e-15
46	1.562e+06	46	0.000e+00
47	1.188e+03	47	0.000e+00
48	6.554e+04	48	0.000e+00
49	9.943e+04	49	0.000e+00
50	0.000e+00	50	-2.842e-14
51	2.172e+04	51	0.000e+00
52	6.291e+04	52	-6.153e+01
53	3.144e+04	53	0.000e+00
54	7.037e+03	54	0.000e+00
55	1.657e+04	55	0.000e+00
56	0.000e+00	56	0.000e+00
57	2.213e+05	57	0.000e+00
58	1.925e+03	58	0.000e+00
59	7.773e+03	59	0.000e+00
60	4.290e+04	60	0.000e+00
61	3.316e+04	61	-4.663e-15

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
62	2.264e+04	62	-9.992e-16
63	9.783e+05	63	0.000e+00
64	7.500e+05	64	-1.770e+00
65	0.000e+00	65	2.665e-15
66	1.000e+06	66	-1.309e+00
67	2.914e+04	67	0.000e+00
68	5.824e+03	68	0.000e+00
69	5.000e+05	69	-5.995e-15
70	0.000e+00	70	0.000e+00
71	0.000e+00	71	-1.794e-13
72	0.000e+00	72	1.332e-14
73	1.476e+04	73	0.000e+00
74	0.000e+00	74	6.395e-14
75	3.736e+03	75	0.000e+00
76	0.000e+00	76	-2.238e-13
77	0.000e+00	77	2.887e-14
78	0.000e+00	78	4.254e-13
79	1.580e+04	79	0.000e+00
80	2.533e+04	80	0.000e+00
81	1.026e+03	81	0.000e+00
82	0.000e+00	82	-1.399e-14
83	0.000e+00	83	-5.085e-14
84	2.223e+03	84	0.000e+00
85	1.132e+04	85	0.000e+00
86	3.339e+05	86	0.000e+00
87	0.000e+00	87	4.441e-15
88	5.912e+05	88	0.000e+00
89	5.323e+05	89	0.000e+00
90	9.092e+03	90	5.329e-15
91	6.506e+04	91	0.000e+00
92	7.596e+04	92	0.000e+00
93	0.000e+00	93	-2.753e-14
94	1.930e+04	94	0.000e+00
95	6.291e+04	95	-6.153e+01
96	3.404e+04	96	-2.065e-13
97	3.208e+04	97	0.000e+00
98	2.948e+04	98	2.145e-13
99	0.000e+00	99	-1.066e-14
100	1.973e+05	100	0.000e+00
101	0.000e+00	101	3.109e-15
102	9.475e+03	102	0.000e+00
103	2.211e+04	103	0.000e+00
104	2.565e+04	104	0.000e+00
105	8.237e+03	105	0.000e+00
106	0.000e+00	106	2.442e-15
107	0.000e+00	107	4.230e+00
108	5.000e+05	108	-3.000e+00
109	1.000e+06	109	-3.309e+00
110	0.000e+00	110	0.000e+00

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
111	0.000e+00	111	0.000e+00
112	0.000e+00	112	0.000e+00
113	1.379e+03	113	0.000e+00
114	4.249e+04	114	0.000e+00
115	5.946e+04	115	0.000e+00
116	7.230e+03	116	0.000e+00
117	4.246e+03	117	0.000e+00
118	1.234e+04	118	0.000e+00
119	1.702e+04	119	2.292e-13
120	0.000e+00	120	0.000e+00
121	6.485e+03	121	0.000e+00
122	1.198e+04	122	0.000e+00
123	8.438e+04	123	0.000e+00
124	1.007e+03	124	0.000e+00
125	8.666e+03	125	-1.059e-13
126	0.000e+00	126	1.876e-13
127	0.000e+00	127	2.220e-15
128	1.132e+04	128	1.121e-14
129	8.336e+04	129	0.000e+00
130	1.100e+06	130	0.000e+00
131	1.472e+05	131	0.000e+00
132	1.033e+06	132	0.000e+00
133	2.375e+03	133	0.000e+00
134	0.000e+00	134	-1.950e-13
135	2.371e+04	135	0.000e+00
136	0.000e+00	136	-3.020e-14
137	0.000e+00	137	4.086e-14
138	6.291e+04	138	-6.153e+01
139	8.785e+03	139	0.000e+00
140	1.210e+04	140	0.000e+00
141	0.000e+00	141	2.105e-13
142	0.000e+00	142	-1.288e-14
143	0.000e+00	143	-1.510e-14
144	0.000e+00	144	-3.109e-15
145	0.000e+00	145	-1.190e-13
146	2.923e+04	146	0.000e+00
147	2.165e+04	147	0.000e+00
148	0.000e+00	148	6.883e-15
149	3.200e+05	149	0.000e+00
150	7.500e+05	150	-1.770e+00
151	1.219e+05	151	0.000e+00
152	1.000e+06	152	-5.309e+00
153	0.000e+00	153	4.174e-14
154	0.000e+00	154	3.020e-14
155	0.000e+00	155	1.665e-14
156	0.000e+00	156	4.885e-14
157	0.000e+00	157	2.167e-13
158	0.000e+00	158	5.862e-14
159	0.000e+00	159	1.332e-13

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
160	0.000e+00	160	0.000e+00
161	0.000e+00	161	5.684e-13
162	0.000e+00	162	2.887e-14
163	0.000e+00	163	3.020e-14
164	4.870e+03	164	0.000e+00
165	8.842e+03	165	0.000e+00
166	0.000e+00	166	2.931e-14
167	7.019e+02	167	0.000e+00
168	6.014e+03	168	0.000e+00
169	0.000e+00	169	5.351e-14
170	0.000e+00	170	1.799e-14
171	5.396e+03	171	0.000e+00
172	0.000e+00	172	9.770e-15
173	8.087e+03	173	0.000e+00
174	1.934e+03	174	0.000e+00
175	1.043e+06	175	0.000e+00
176	2.832e+03	176	0.000e+00
177	3.375e+04	177	0.000e+00
178	4.894e+03	178	0.000e+00
179	2.130e+04	179	0.000e+00
180	8.243e+03	180	0.000e+00
181	6.291e+04	181	-6.153e+01
182	2.626e+04	182	0.000e+00
183	1.675e+04	183	0.000e+00
184	0.000e+00	184	1.883e-13
185	0.000e+00	185	-3.819e-14
186	8.645e+04	186	0.000e+00
187	0.000e+00	187	-1.288e-14
188	0.000e+00	188	-1.275e-13
189	1.247e+04	189	0.000e+00
190	1.512e+04	190	0.000e+00
191	1.242e+04	191	0.000e+00
192	2.667e+05	192	0.000e+00
193	0.000e+00	193	4.230e+00
194	0.000e+00	194	3.000e+00
195	1.000e+06	195	-3.309e+00
196	0.000e+00	196	-1.776e-14
197	0.000e+00	197	-3.020e-14
198	0.000e+00	198	-1.132e-14
199	0.000e+00	199	-4.086e-14
200	0.000e+00	200	-1.155e-13
201	0.000e+00	201	-6.217e-14
202	9.137e+03	202	0.000e+00
203	0.000e+00	203	-5.684e-14
204	2.391e+04	204	0.000e+00
205	1.702e+04	205	-3.197e-14
206	0.000e+00	206	-6.617e-14
207	0.000e+00	207	-3.535e-13
208	6.097e+03	208	0.000e+00

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
209	0.000e+00	209	-3.304e-13
210	4.573e+02	210	0.000e+00
211	3.849e+03	211	0.000e+00
212	0.000e+00	212	-3.999e-13
213	0.000e+00	213	-2.220e-14
214	7.732e+03	214	0.000e+00
215	0.000e+00	215	3.575e-14
216	1.144e+06	216	0.000e+00
217	1.567e+05	217	0.000e+00
218	1.035e+06	218	0.000e+00
219	6.943e+03	219	0.000e+00
220	4.642e+04	220	0.000e+00
221	1.040e+05	221	0.000e+00
222	0.000e+00	222	8.882e-15
223	1.562e+04	223	0.000e+00
224	6.291e+04	224	-6.153e+01
225	1.271e+04	225	0.000e+00
226	2.473e+04	226	0.000e+00
227	1.319e+04	227	0.000e+00
228	0.000e+00	228	3.419e-14
229	1.462e+05	229	0.000e+00
230	0.000e+00	230	-1.421e-14
231	0.000e+00	231	-1.150e-13
232	3.912e+04	232	0.000e+00
233	2.895e+04	233	0.000e+00
234	0.000e+00	234	2.276e-15
235	4.839e+05	235	0.000e+00
236	7.500e+05	236	-1.770e+00
237	1.288e+05	237	0.000e+00
238	1.000e+06	238	-1.309e+00
239	0.000e+00	239	4.230e+00
240	0.000e+00	240	3.208e+00
241	0.000e+00	241	1.359e+00
242	0.000e+00	242	4.937e+00
243	0.000e+00	243	2.651e+00
244	0.000e+00	244	3.358e+00
245	0.000e+00	245	1.004e+01
246	0.000e+00	246	7.054e+00
247	0.000e+00	247	6.939e+01
248	0.000e+00	248	2.712e+00
249	0.000e+00	249	2.654e+00
250	0.000e+00	250	3.920e+00
251	0.000e+00	251	4.135e+00
252	0.000e+00	252	4.494e+00
253	0.000e+00	253	9.970e-01
254	0.000e+00	254	8.509e-01
255	0.000e+00	255	1.147e+00
256	2.702e+03	256	0.000e+00
257	0.000e+00	257	5.254e-01

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
258	0.000e+00	258	8.912e-01
259	6.428e+04	259	0.000e+00
260	5.987e+05	260	0.000e+00
261	1.040e+06	261	0.000e+00
262	6.246e+03	262	0.000e+00
263	4.162e+04	263	0.000e+00
264	1.164e+04	264	0.000e+00
265	1.390e+04	265	0.000e+00
266	1.234e+04	266	0.000e+00
267	6.291e+04	267	-1.168e+01
268	1.899e+04	268	0.000e+00
269	3.349e+04	269	0.000e+00
270	1.636e+04	270	0.000e+00
271	6.697e+03	271	0.000e+00
272	1.573e+05	272	0.000e+00
273	3.736e+02	273	0.000e+00
274	3.418e+03	274	0.000e+00
275	2.385e+04	275	0.000e+00
276	3.316e+04	276	-7.624e-01
277	1.156e+04	277	0.000e+00
278	4.016e+05	278	0.000e+00
279	3.504e+04	279	0.000e+00
280	5.000e+05	280	-6.208e+00
281	1.000e+06	281	-6.684e-01
282	0.000e+00	282	1.000e+00
283	0.000e+00	283	7.921e-01
284	0.000e+00	284	3.316e-01
285	0.000e+00	285	1.210e+00
286	0.000e+00	286	6.328e-01
287	0.000e+00	287	7.993e-01
288	0.000e+00	288	2.372e+00
289	0.000e+00	289	1.677e+00
290	0.000e+00	290	1.628e+01
291	0.000e+00	291	6.512e-01
292	0.000e+00	292	6.576e-01
293	0.000e+00	293	9.620e-01
294	0.000e+00	294	9.868e-01
295	0.000e+00	295	1.071e+00
296	0.000e+00	296	2.427e-01
297	0.000e+00	297	2.055e-01
298	0.000e+00	298	2.778e-01
299	0.000e+00	299	1.076e+00
300	0.000e+00	300	1.317e-01
301	0.000e+00	301	2.216e-01

3 Variáveis Duais (Multiplicadores de Lagrange)

Tabela 2: Variáveis duais (y) do problema grow7

Coordenada y	Valor y (Dual)
1	5.230e+00
2	4.000e+00
3	1.691e+00
4	6.147e+00
5	3.284e+00
6	4.158e+00
7	1.242e+01
8	8.731e+00
9	8.567e+01
10	3.364e+00
11	3.312e+00
12	4.882e+00
13	5.122e+00
14	5.564e+00
15	1.240e+00
16	1.056e+00
17	1.425e+00
18	1.076e+00
19	6.571e-01
20	1.113e+00
21	5.230e+00
22	4.000e+00
23	1.691e+00
24	6.147e+00
25	3.284e+00
26	4.158e+00
27	1.242e+01
28	8.731e+00
29	8.567e+01
30	3.364e+00
31	3.312e+00
32	4.882e+00
33	5.122e+00
34	5.564e+00
35	1.240e+00
36	1.056e+00
37	1.425e+00
38	1.076e+00
39	6.571e-01
40	1.113e+00
41	5.230e+00
42	4.000e+00
43	1.691e+00
44	6.147e+00
45	3.284e+00
46	4.158e+00
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Coordenada y	Valor y (Dual)
47	1.242e+01
48	8.731e+00
49	8.567e+01
50	3.364e+00
51	3.312e+00
52	4.882e+00
53	5.122e+00
54	5.564e+00
55	1.240e+00
56	1.056e+00
57	1.425e+00
58	1.076e+00
59	6.571e-01
60	1.113e+00
61	5.230e+00
62	4.000e+00
63	1.691e+00
64	6.147e+00
65	3.284e+00
66	4.158e+00
67	1.242e+01
68	8.731e+00
69	8.567e+01
70	3.364e+00
71	3.312e+00
72	4.882e+00
73	5.122e+00
74	5.564e+00
75	1.240e+00
76	1.056e+00
77	1.425e+00
78	1.076e+00
79	6.571e-01
80	1.113e+00
81	5.230e+00
82	4.000e+00
83	1.691e+00
84	6.147e+00
85	3.284e+00
86	4.158e+00
87	1.242e+01
88	8.731e+00
89	8.567e+01
90	3.364e+00
91	3.312e+00
92	4.882e+00
93	5.122e+00
94	5.564e+00
95	1.240e+00
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Coordenada y	Valor y (Dual)
96	1.056e+00
97	1.425e+00
98	1.076e+00
99	6.571e-01
100	1.113e+00
101	5.230e+00
102	4.000e+00
103	1.691e+00
104	6.147e+00
105	3.284e+00
106	4.158e+00
107	1.242e+01
108	8.731e+00
109	8.567e+01
110	3.364e+00
111	3.312e+00
112	4.882e+00
113	5.122e+00
114	5.564e+00
115	1.240e+00
116	1.056e+00
117	1.425e+00
118	1.076e+00
119	6.571e-01
120	1.113e+00
121	1.000e+00
122	7.921e-01
123	3.316e-01
124	1.210e+00
125	6.328e-01
126	7.993e-01
127	2.372e+00
128	1.677e+00
129	1.628e+01
130	6.512e-01
131	6.576e-01
132	9.620e-01
133	9.868e-01
134	1.071e+00
135	2.427e-01
136	2.055e-01
137	2.778e-01
138	1.076e+00
139	1.317e-01
140	2.216e-01